Ahmet Emir Kalafat

Computer Engineer, Electrical and Electronics Engineer

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ABOUT ME

With a primary degree in Electrical and Electronics Engineering, complemented by a double major in Computer Engineering, I have built a unique skill set that bridges hardware principles with advanced software development. My career began at SameUp, where I honed my systematic problem-solving and collaborative skills by contributing to two mobile applications using React Native. This role provided deep insights into project lifecycle and effective teamwork.

Currently, at Talsen, I am developing a complex LLM code-writing infrastructure within a rigorous agile framework. Leading the frontend and CLI structure for the ARA Project Management Program has sharpened my ability to design, develop, and integrate complex systems while collaborating with an international team. I am now eager to leverage this powerful combination of core engineering knowledge and cutting-edge software expertise to tackle new challenges and contribute to innovative projects in the electrical engineering field.

EDUCATION

Computer Engineering (Double Major)
Fatih Sultan Mehmet Foundation University - İstanbul
2021 - 2025

Electrical-Electronics Engineering (GPA: 3.0)
Fatih Sultan Mehmet Foundation University - İstanbul
2020 - 2024

Electrical and Automation Engineering (Erasmus+)
Klaipeda Valstybinė Koleija - Lithuania
2023 January – 2023 June

EXPERIENCE

Talsen Team GmbH - Software Developer
2024 July - Current

I joined the Talsen Team as a part-time job while I was still studying. I played a major role in the ARA Project Management Software that the team created. I worked on both the frontend and the cli part of the project. We used React for frontend and Python for the ara-cli project. Here in the Talsen Team we are strictly using LLM for better productivity. The ARA infrastructure we created is fully integrated with the assistance of LLM agents.

SameUp - Mobile Application Developer
2023 July - 2024 July

I joined SameUp as an intern while I was still studying. After my compulsory internship ended, I continued to work part-time. I played a pivotal role in the UI/UX overhaul of the Kidokit application by implementing a new design language across more than 20 core screens using React Native. Co-developed the Investag investment tracking application from the ground up, personally architecting and building over 14 key features and reusable components with React Native. I also worked on the Teknosa website frontend with the PUG framework for project prototyping for next updates.

SKILLS

MatLab
Raspberry Pi
C (Embedded Systems), C#
STM32 (Embedded Processor)
AWS
CoDeSys (FBD, IL, LD, ST+)
React-Native (iOS-Android)
React
JavaScript
HTML, CSS
pug
Git, GitHub
Flutter (Riverpod, Provider)
Firebase (Storage, Auth, Firestore, Cloud Messaging)
Java
Python (pydantic, pydantic_ai, litellm, pandas, OpenCV, Tkinter, pygame)

LANGUAGES

Turkish - Native
English - C1 (Advanced)

COMMUNITIES

New Generation R&D - Board Member

My University Lecture Notes Mobile Application - Moderator

GRADUATION PROJECT

Barcode Reading System with Microprocessor

I successfully completed my undergraduate graduation project, focusing on the design and implementation of a camera-assisted barcode reading technology aimed at identifying and reliably reading product barcodes. I developed this system using the Raspberry Pi 4 microprocessor module and the OV5647 5 Megapixel camera module. My methodology employed advanced image processing via OpenCV, utilizing techniques such as Morphological processes (Erosion and Dilate) and Thresholding to ensure the acquisition of clean, noise-free barcode images. To minimize errors from incorrect exposures, I specifically integrated a user-controlled photo taking function. The processed pixel data was meticulously configured to comply with the EAN-13 barcode standard, demonstrating an efficient and fast solution suitable for applications in sectors like retail, logistics, and production tracking

PERSONAL PROJECTS

Vehicle Controlled via Bluetooth

Collaborated in a team of three to design, build, and program a smart vehicle controlled via a custom mobile application. Programmed an STM32 processor using **C** to manage all core functionalities, including real-time motor control, 360-degree rotation, and processing sensor data. Implemented a robust safety feature using front and rear sensors to automatically detect obstacles and override user commands to prevent collisions. developed the control application using **Flutter**, creating an intuitive user interface for seamless vehicle operation via Bluetooth communication. Successfully integrated the embedded hardware with the mobile application, ensuring reliable and responsive command execution for features like remote headlight activation and precise movement.

Python Platform Game

Platform game project using Python Tkinter and pygame libraries. This project allowed me to learn how to create interfaces in Python, save information locally, and read saved information.
Project Link: https://github.com/emirkalafat/python_game

Food Storage -Flutter Mobile Application

I created an application with Flutter where users can share their own recipes, photos of the dishes and the ingredients they use. Features of the application: I made an application where people can like other people's recipes and follow people. While making this application, I used the Riverpod library for "State Management". I used Google Firebase to store user and recipe data. The app is currently available on Play Store. Published on TestFlight for iOS.